

This is a **real-world Matplotlib Data Visualization Project Assignment** that covers all the topics from Intro to Pie Charts. You will not only learn plotting but also develop **business insights** from data, just like a Data Analyst.

Project Title

E-Commerce Sales Analysis Dashboard using Matplotlib

Objective

Analyze an e-commerce company's sales data and generate various visualizations to answer business questions.

Dataset

Create a CSV file named:

ecommerce_sales.csv

OrderID, Month, Sales, Profit, Customers, Orders, Category

1001, Jan, 45000, 8000, 120, 150, Electronics
1002, Feb, 48000, 8500, 130, 160, Electronics
1003, Mar, 55000, 9500, 140, 175, Electronics
1004, Apr, 60000, 11000, 155, 190, Electronics
1005, May, 68000, 13000, 170, 210, Electronics
1006, Jun, 72000, 14500, 185, 225, Electronics
1007, Jul, 78000, 16000, 200, 245, Electronics
1008, Aug, 82000, 17500, 215, 260, Electronics
1009, Sep, 79000, 16800, 210, 250, Electronics
1010, Oct, 85000, 18500, 225, 275, Electronics
1011, Nov, 92000, 20500, 240, 295, Electronics
1012, Dec, 105000, 24000, 275, 330, Electronics

Part 1 – Basic Plotting

Task 1

Create a line plot of:

- X-axis → Month
- Y-axis → Sales

Questions

1. Which month recorded maximum sales?
2. Which month recorded minimum sales?
3. Is the overall sales trend increasing or decreasing?

Part 2 – Markers and Line Styles

Task 2

Create a line chart:

- Marker = 'o'
- Line Color = Green
- Line Style = '--'

Plot Sales vs Month.

Questions

1. Between which months is the highest jump in sales?
 2. During which month did sales decline?
-

Part 3 – Labels and Grid

Task 3

Create a chart showing Profit vs Month.

Requirements:

- Title
- X Label
- Y Label
- Grid Enabled

Questions

1. What is the total annual profit?
 2. Which month generated the highest profit?
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Part 4 – Scatter Plot Analysis

Task 4

Create a scatter plot:

- X-axis = Customers
- Y-axis = Sales

Questions

1. Is there a relationship between customers and sales?
 2. Does increasing customers increase sales?
 3. Identify any outlier points.
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Part 5 – Bar Chart Analysis

Task 5

Create a bar chart showing Orders by Month.

Questions

1. Which month had the highest orders?
 2. Which month had the lowest orders?
 3. What is the difference between highest and lowest order counts?
-

Part 6 – Histogram Analysis

Task 6

Create a histogram of Sales.

Use:

```
plt.hist(df['Sales'], bins=5)
```

Questions

1. Which sales range occurs most frequently?
 2. Are sales values evenly distributed?
 3. Is the distribution skewed?
-

Part 7 – Pie Chart Analysis

Category Sales Data

Category, Sales

Electronics, 450000

Fashion, 320000

Grocery, 280000

Furniture, 180000

Sports, 120000

Task 7

Create a Pie Chart.

Questions

1. Which category contributes the most sales?

2. Which category contributes the least sales?
3. What percentage of total sales comes from Electronics?

Part 8 – Subplots (Mini Dashboard)

Create a dashboard with 2×2 subplots.

Chart 1

Sales Trend

Chart 2

Profit Trend

Chart 3

Orders Bar Chart

Chart 4

Customer vs Sales Scatter Plot

Questions

1. Which chart provides the most useful business insight?
2. Are sales and profit moving together?
3. Do more customers always lead to more sales?

Part 9 – Business Analysis Challenge

Using all visualizations, answer:

Sales Performance

1. Best performing month?
2. Worst performing month?
3. Sales growth from Jan to Dec?

Profit Analysis

4. Total annual profit?
5. Average monthly profit?

Customer Analysis

6. Average customers per month?
7. Peak customer month?

Order Analysis

8. Average monthly orders?
9. Peak order month?

Category Analysis

10. Highest revenue category?
11. Lowest revenue category?

Part 10 – Final Professional Dashboard

Create a dashboard containing:

- Sales Line Chart
- Profit Line Chart
- Orders Bar Chart
- Customer Scatter Plot
- Category Pie Chart

Requirements:

- ✓ Figure Size = (15,10)
 - ✓ Titles for every chart
 - ✓ Grid where applicable
 - ✓ Different colors
 - ✓ Proper labels
 - ✓ Tight Layout
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Deliverables

Students must submit:

1. Python Script (.py)
2. CSV Dataset
3. Dashboard Screenshot
4. Business Insights Report (minimum 10 insights)

This assignment simulates a real Data Analyst project and gives hands-on practice with every major Matplotlib visualization type.

